BREATH TRAINING

A smart breathing trainer and its companion app

UX/UI design · Visual design · Illustration · Product design
Solo project · Design workshop with an instructor · Jun – Aug 2020

Jie (Fia) Yin

UX/UI · VISUAL DESIGN

Why

After COVID-19, many recovered patients leave hospital with reduced lung function and need breathing training at home.

What

A portable breathing trainer plus an app for lung-capacity tests, solo and multiplayer training games, a friends leaderboard, hospital services, disinfection booking and device rental.

Design focus

From user research and information architecture to wireframes and hi-fi UI, with one visual language across the app, the illustrations and the game world. Product form modeled in Rhino.

Inside

Background and SET analysis · Persona and insights · Empathy map and competitors · User journey · Stakeholder map and information architecture · Sketches · Product design · User flow · Visual showcase

BREATH TRAINING · UX/UI & VISUAL DESIGN

Gamified breathing training for lung recovery at home

A companion app for a smart breathing trainer: device pairing, map-based training games, community and device care.

COLOR

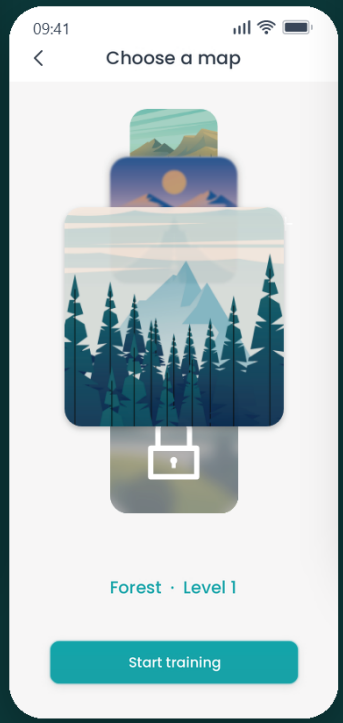
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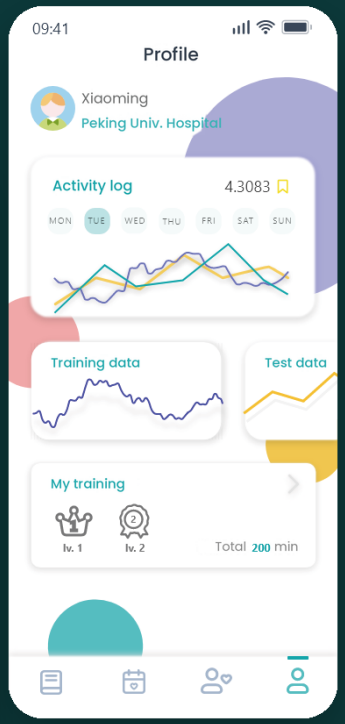
Training game · Forest, level 1



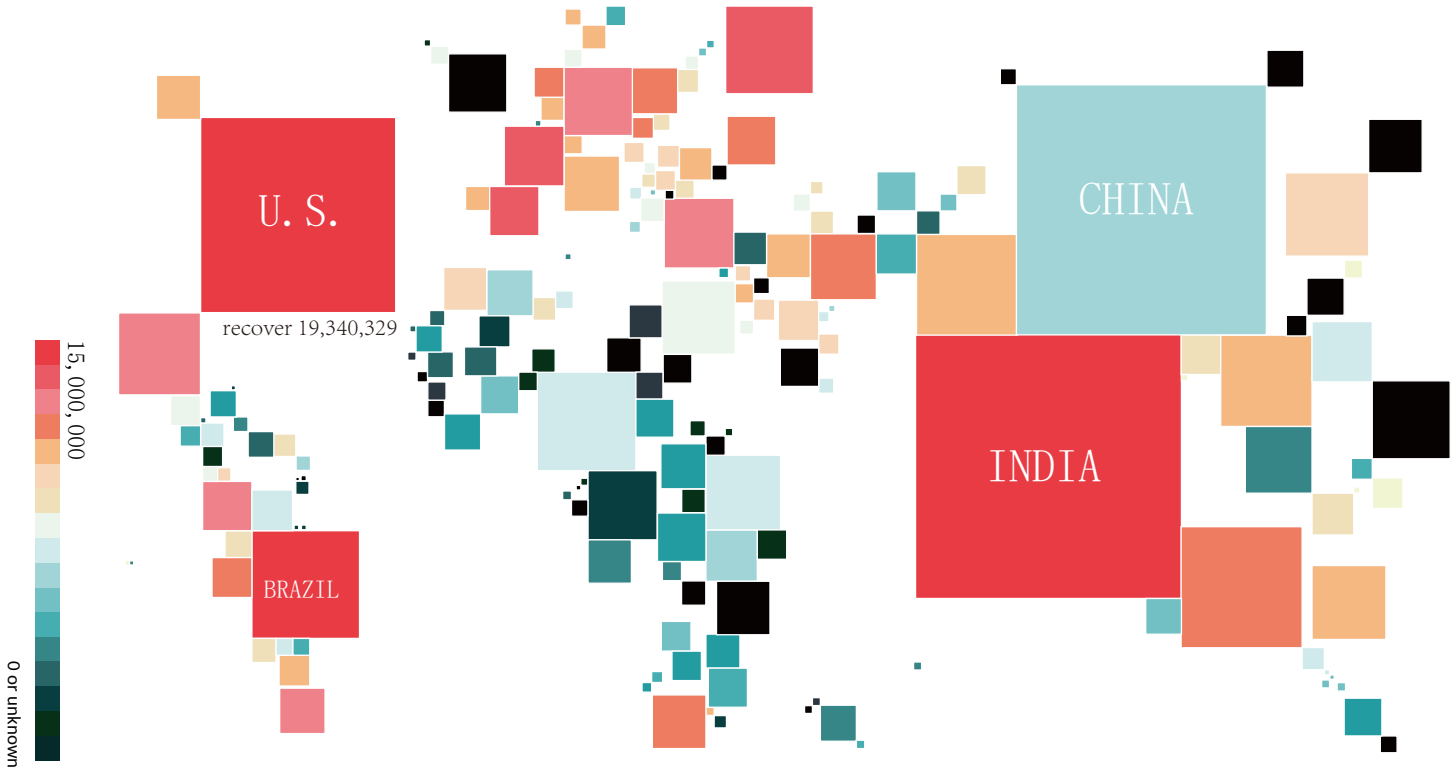
Home
Device status at a glance



Choose a map
Pick a scene and a level



Profile & progress
Records, data and badges



data recourse: WHO Coronavirus Disease (COVID-19) Dashboard 2021.2.29

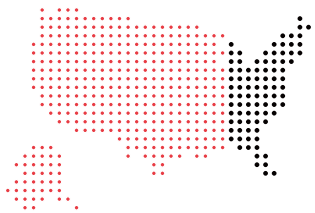
Background

COVID-19 began spreading in December 2019. For hospitalized non-severe patients, doctors recommend lung breathing trainers, which play a major role around surgery. As countries brought the pandemic under control, recoveries kept rising: 65,541 in China alone and 19,340,329 in the U.S. As these patients leave hospital, their lasting after-effects caught our attention, and breathing trainers also play a key role in prescribed rehabilitation.



101,796—4,843=96,953

95%



28,974,623—518,363=28,456,260

98%



4,144,577—121,747=4,022,830

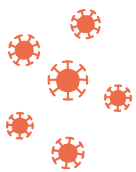
Cumulative cases – cumulative deaths = potential users

Background





Since December 2019, the world has faced an unprecedented crisis: the COVID-19 pandemic. Prevention and control in China have improved, and treatment for COVID-19 pneumonia has matured. But as patients recover and leave hospital, a new problem appears: the virus can seriously damage lung function, and even patients who meet the clinical cure standard may still have weak breathing and reduced physical activity after discharge, needing a long recovery. Medical devices are therefore needed to help recovering patients train their lungs and regain their health.



COVID-19





Paralysis



Weak breathing



Reduced mobility



Social difficulties



Mental health issues

SET

SOCIAL

With mature prevention and treatment, China has seen nearly 100,000 recoveries. All of these patients need lung rehabilitation, both in hospital and after discharge, so the market is large.

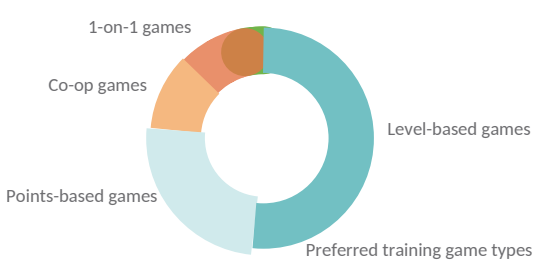
ECONOMIC

China's economy has kept growing steadily after the pandemic, and its industries are recovering. The crisis also brings opportunity: the medical industry, for example, keeps growing.

TECHNOLOGICAL

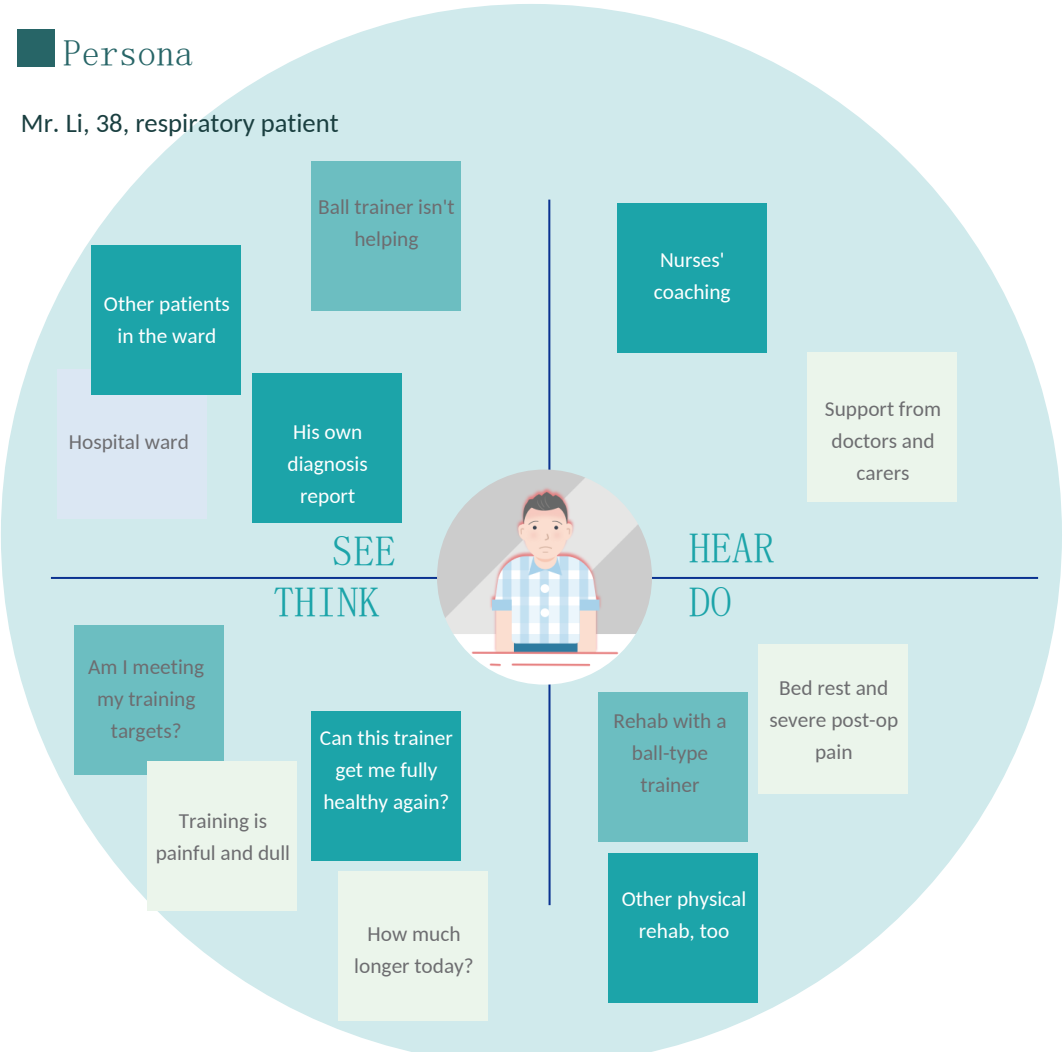
Existing trainers use simple principles (pressure difference, gas expansion, friction, a narrower mouthpiece) to exercise the lungs, so the technology is easy to build.





Persona

Mr. Li, 38, respiratory patient



Insight

PAIN POINTS

Can only judge lung function roughly by how many balls rise;

Training is long and boring;

Trainers can only be used in hospital;

HOPES&NEEDS

A clearer, precise trainer that measures, shows and records lung recovery;

More engaging training that keeps patients positive and speeds recovery;

Want to keep training after discharge;

IDEATIONS

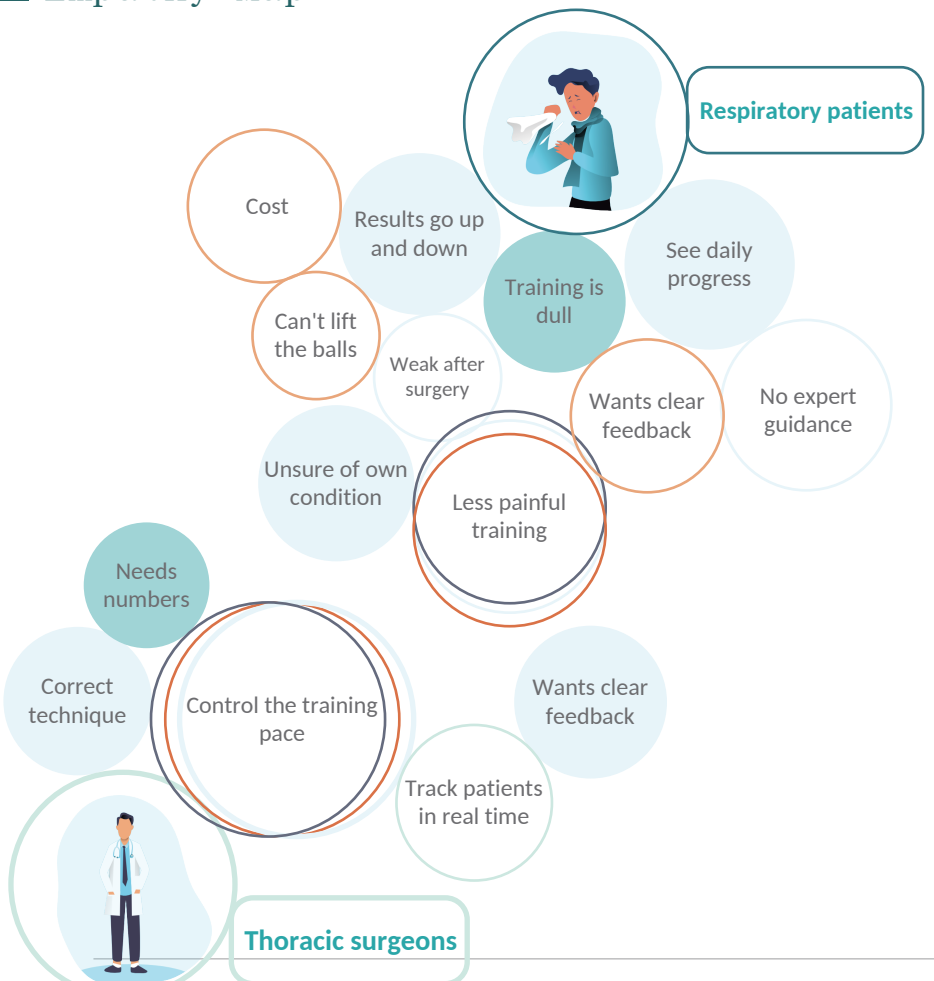
Easy, effective lung rehab training

An app that stores and shows training data

Gamified post-surgery rehab training

Break lung recovery into measurable steps

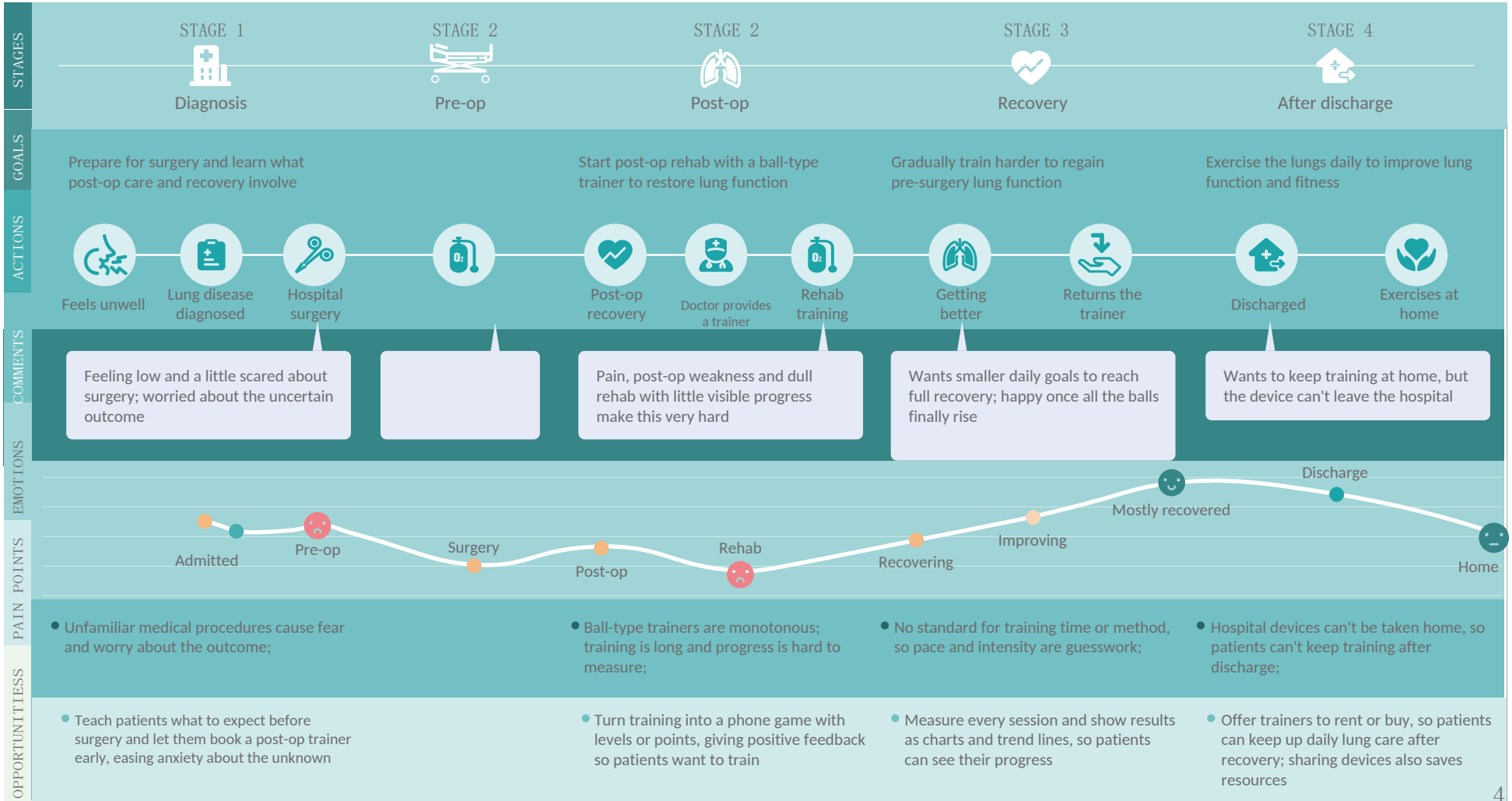
Empathy Map



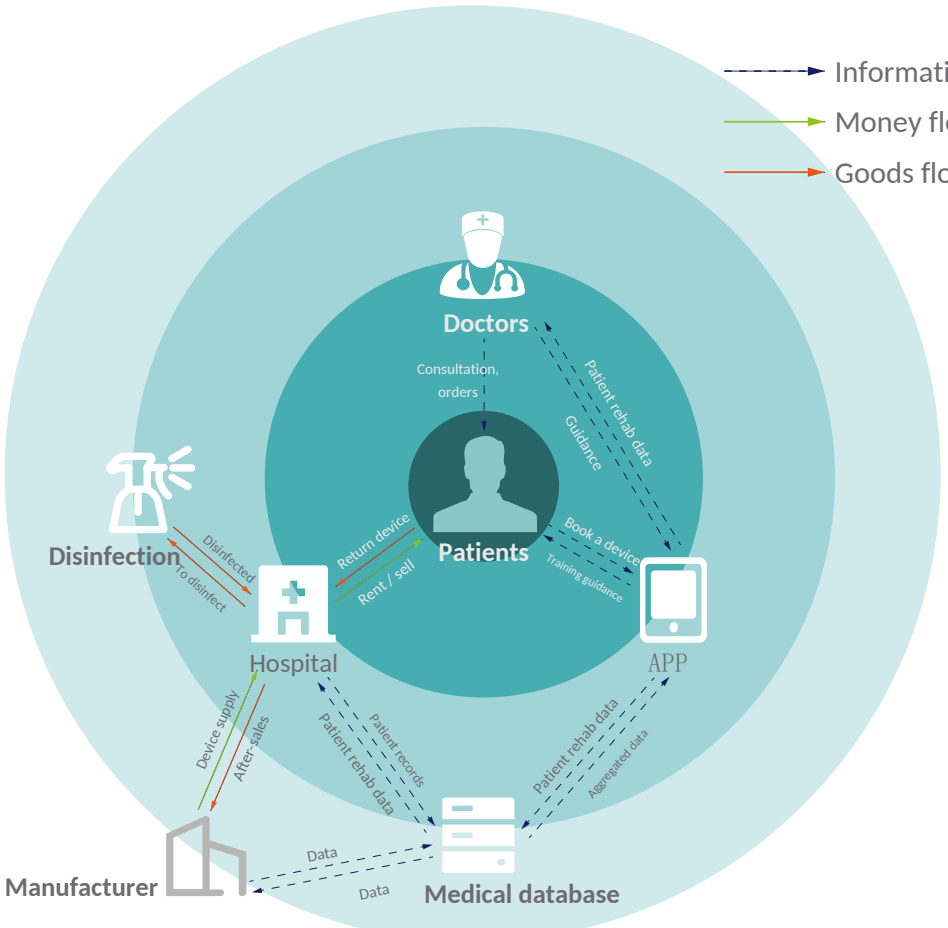
Competitive Products

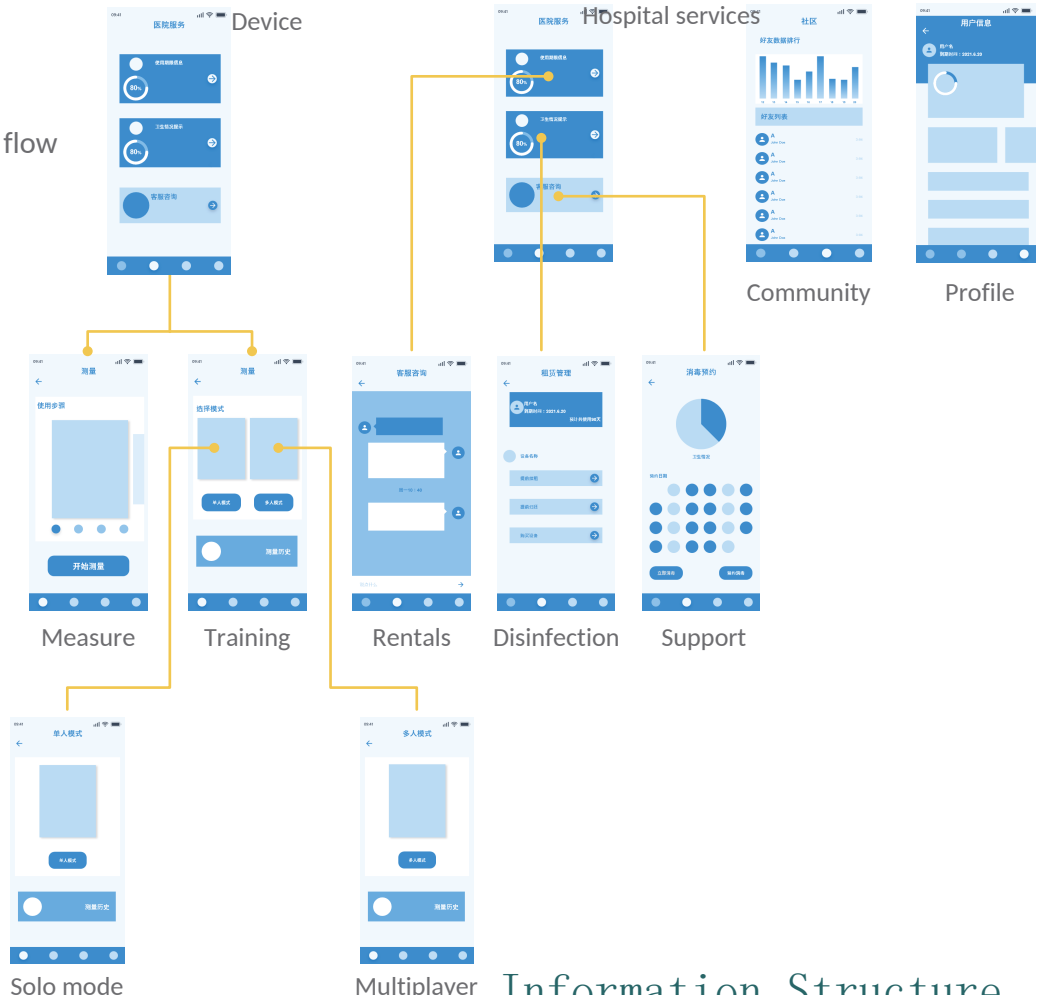
PRODUCTS	 Jianyanshe lung trainer	 Sonmol breathing trainer	 KOO breathing trainer
PROFILE	• Trains breathing muscles; • Strengthens the lungs; • Portable mask-style unit;	• Hospital-only; • Simple, easy to use; • Adjustable resistance;	• Inhale and exhale modes; • Graduated resistance; • Builds strength, clears mucus;
PRICE	¥228	¥419	¥208
MATERIALS	Translucent white plastic; Mixed fabrics, etc.;	Medical rubber; Medical plastic, etc.;	Mainly eco-friendly medical plastic;
PROS	• Selectable modes; • Gradual training, rising difficulty;	• Video tutorials included; • Trains inhale and exhale; • Small, portable, washable;	• No need to flip it for exhale practice; • Valve-set difficulty; • Well-reviewed brand;
CONS	• Mask is uncomfortable; • Stiff design; • No visible progress;	• Pricey; • Fixed mouthpiece;	• Moisture builds up inside; • Balls get stuck;

User Journey Map



Stakeholders Map



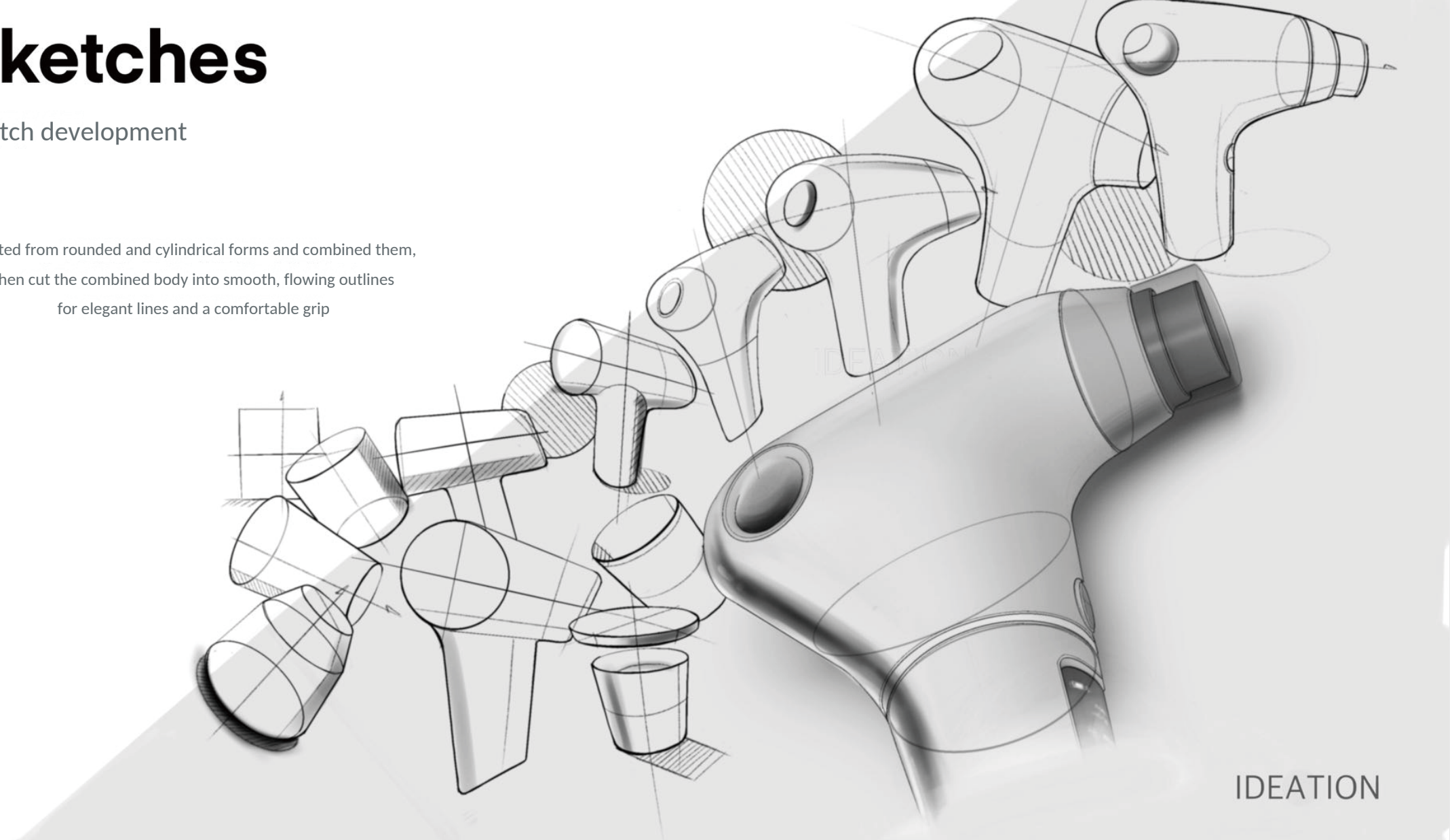


Information Structure

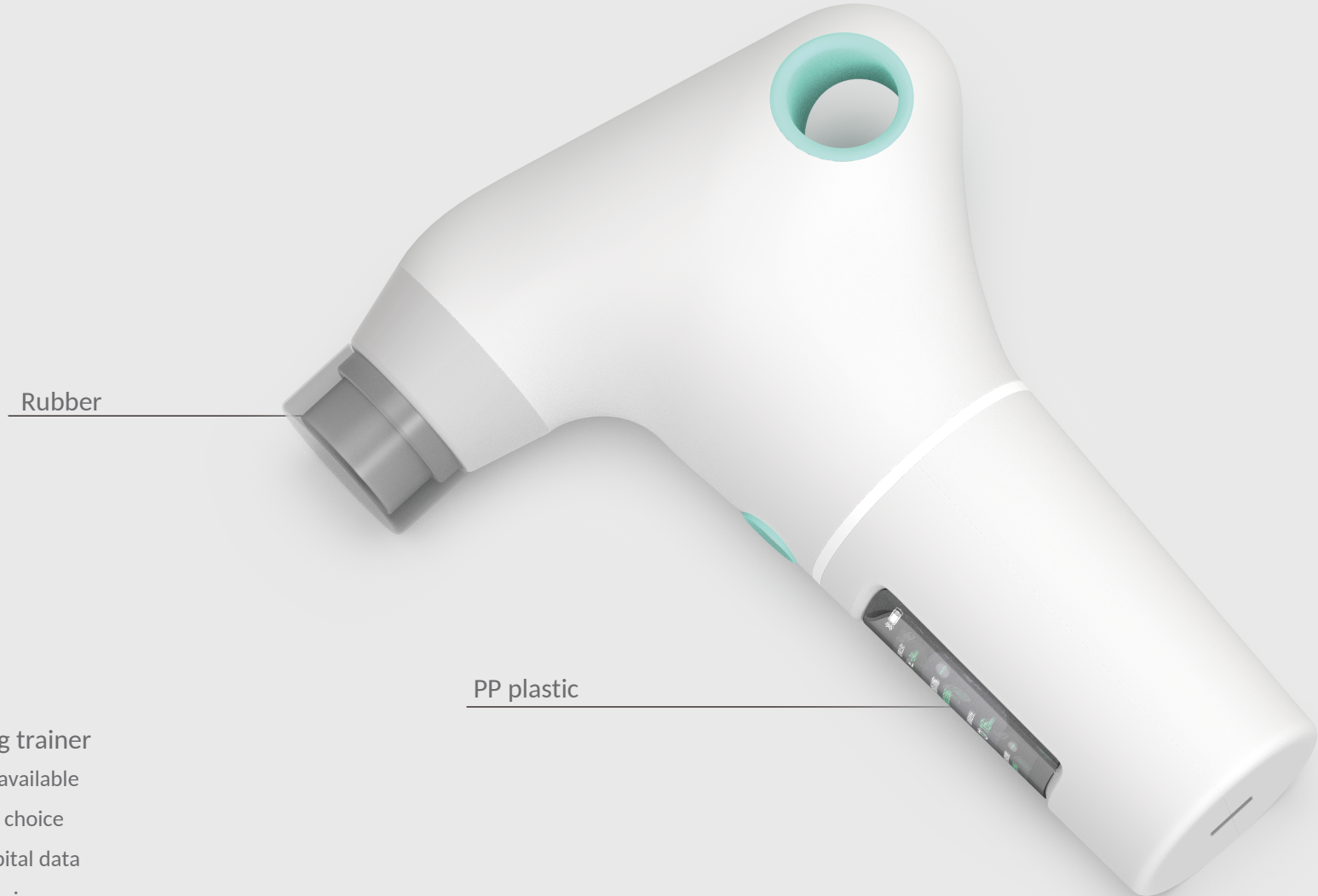
Sketches

Sketch development

Started from rounded and cylindrical forms and combined them,
then cut the combined body into smooth, flowing outlines
for elegant lines and a comfortable grip



DEVELOPMENT



Smart breathing trainer

Short-term rental available

A more affordable choice

Connected to hospital data

More attentive service

BREATH TRAINING

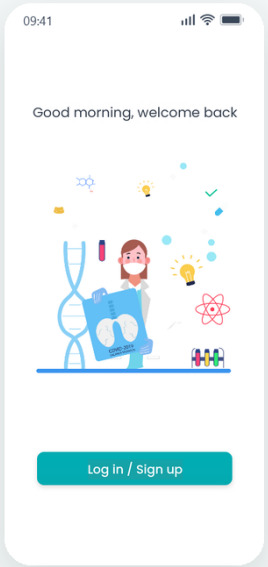
Gamified breathing training for lung recovery at home

UX/UI and visual design for a smart breathing trainer's companion app: wireframes, hi-fi screens, illustrations and the training game.

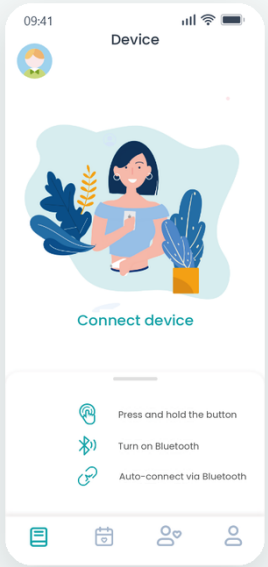
Jie (Fia) Yin · 2020



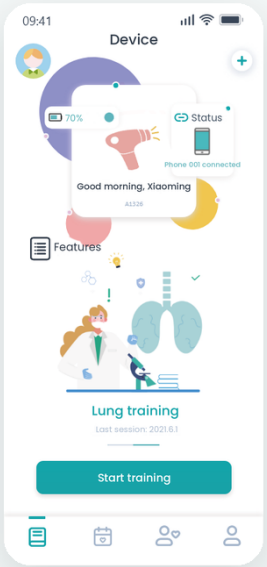
01 Welcome



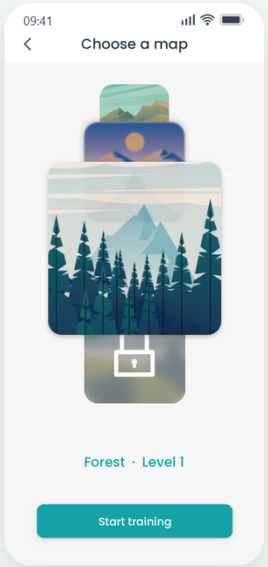
02 Pair the device



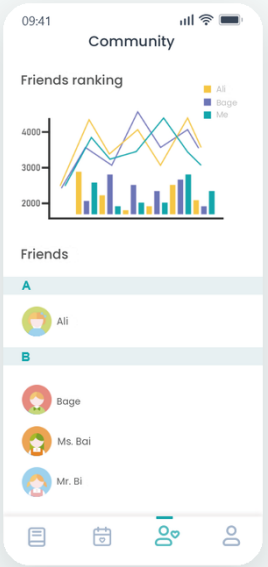
03 Home



04 Choose a map



05 Community



06 Hygiene & booking

